AFRICAN DEVELOPMENT BANK

DATA INNOVATION FOR AFRICA (DI4A)

Night-time lights in practice

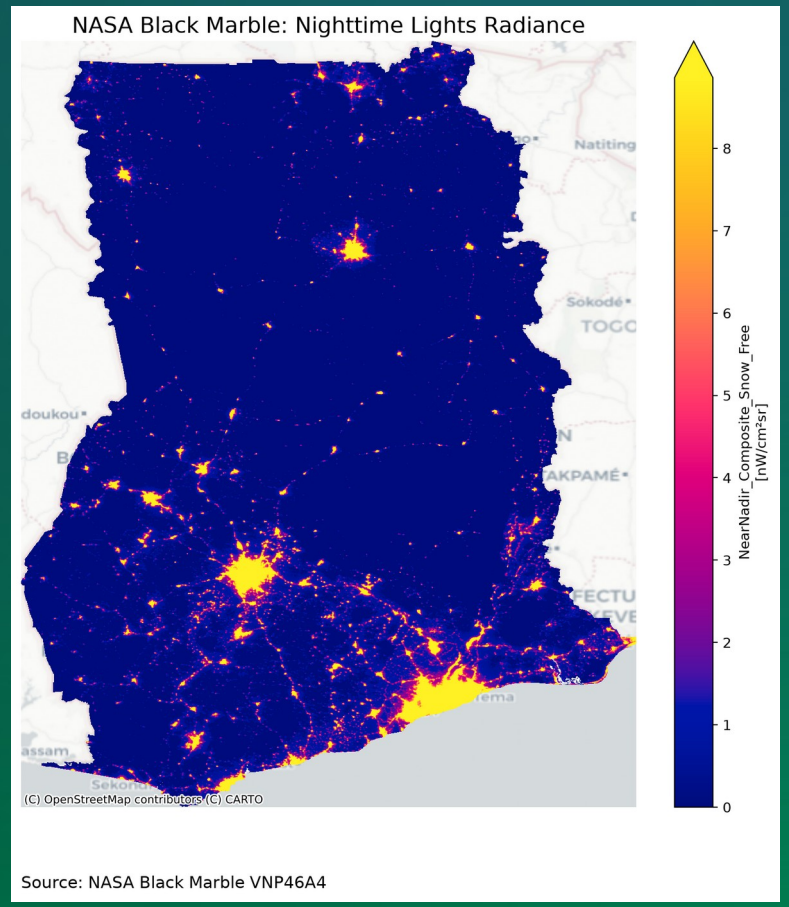
What they have made it possible to measure — from satellite to published dashboard

Analysis notebook “NTL hands-on, part 3”: a reproducible pipeline driven by one ISO3 code

AFDB – DATA INNOVATION FOR AFRICA (DI4A)

Hands-on series · Satellite data for economic statistics

Part 3 · Analyse and publish



Black Marble VNP46A4 (2025), Ghana — real image

Source: NASA Black Marble via blackmarblepy (World Bank, MIT licence)

By the end of this session, you will be able to…

Explain

what night-time lights measure, why they inform on economic activity, and where their limits lie.

Reproduce

the end-to-end pipeline: one ISO3 code → national and regional indicators → a public dashboard.

Interpret

the key metrics: STL, Theil–Sen, convergence, NTL–GDP elasticity, shocks, Gini, population “in the dark”.

Publish with care

results that carry a confidence level, intervals and full traceability.

Common thread: *a measurement is only useful if you can say how far it can be trusted.*

Session outline (75 minutes)

10 min	A	Why lights measure the economy
10 min	B	Sensors and data
15 min	C–D	Architecture, pipeline and preprocessing
25 min	E	Analyses: trends, regions, GDP, shocks, inequality
7 min	F	Confidence, dashboard and publishing
8 min	G	Limitations, wrap-up and bridge to your projects

Where we are

Parts 1–2

Collect and understand a Black Marble file

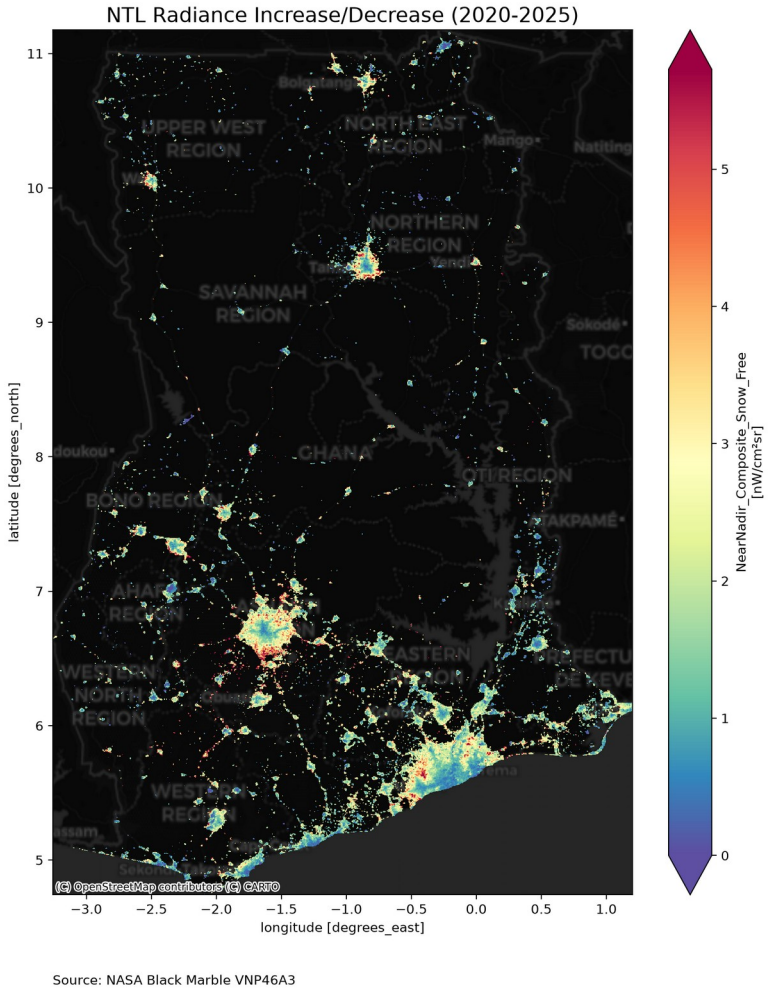
Part 3

Analyse and publish — this session

Next step

Apply the pipeline to your own country

The intuition: human activity is visible from space



Radiance change 2020→2025, Ghana (Black Marble). Source: blackmarblepy, World Bank (MIT).

Activity

producing, transporting, trading, consuming

Energy

activity requires electricity

Light

part of it escapes skywards at night

Measurement

the VIIRS sensor records it every night

Homogeneous

same sensor, same method, everywhere

Detailed

~500 m pixel: regions, districts, cities

Timely

monthly or even daily, without waiting for the accounts

Why it matters national accounts are annual, published late and rarely regional; lights fill these gaps without replacing official statistics.

What the literature says

2011 — **Chen & Nordhaus** · *PNAS*

Lights add information mainly where statistics are weak, and for regions.

2012 — **Henderson, Storeygard & Weil** · *AER*

Growth in lights tracks GDP growth; they improve growth estimates.

2017–21 — **Elvidge et al.** · *IJRS · RS*

Monthly and annual VIIRS products (VNL V2): stray light corrected, background and ephemeral lights masked.

2018 — **Román et al.** · *RSE*

NASA Black Marble: BRDF and lunar corrections, suited to daily series.

2020 — **Li et al.** · *Sci. Data*

Harmonised DMSP–VIIRS series 1992–2018: the long run, to be read as a trend.

2020–21 — **Gibson et al.** · *JDE · JES*

Limits in rural and sparsely populated areas: the signal is weak there.

≈ 0.3

elasticity of GDP with respect to lights (Henderson et al., 2012): +1% light ↔ ≈ +0.3% GDP, on average across a global panel

Key point

This figure is an international average. Your country will have its own elasticity, estimated on only 10 to 30 years: it will be more uncertain. The notebook estimates it and shows its interval.

What lights see… and what they miss

Satellite overpass around 01:30 local time. Lights measure night-time radiance, not value added.

✓ Well captured

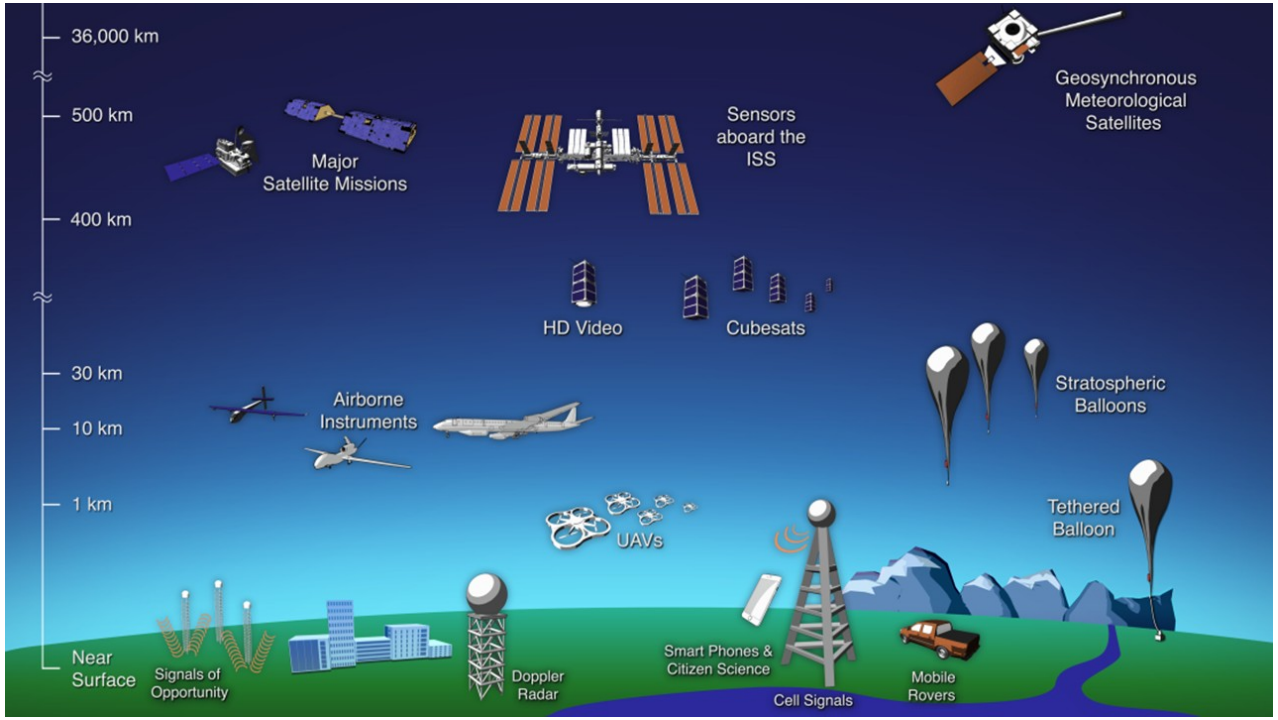
- Cities, street lighting, commercial areas
- Industry, ports, mines, energy sites
- Transport corridors, new electrification

× Poorly captured or misleading

- Agriculture and the rural informal economy
- Daytime trade and dimly lit services
- Gas flares, fires, aurora: light without local activity

Consequence a country whose growth comes from agriculture or daytime services can grow without its light moving. Notebook section 10 and the dashboard's "Methods & limitations" page detail these biases.

How a satellite sees at night



Observation platforms, from ground to geostationary orbit. Source: NASA Goddard, via Open Nighttime Lights (World Bank, CC BY 4.0).

Suomi-NPP · NOAA-20

sun-synchronous polar orbit, ~830 km altitude

VIIRS sensor, DNB band

Day/Night Band: sensitive to very faint night-time light

One overpass per night

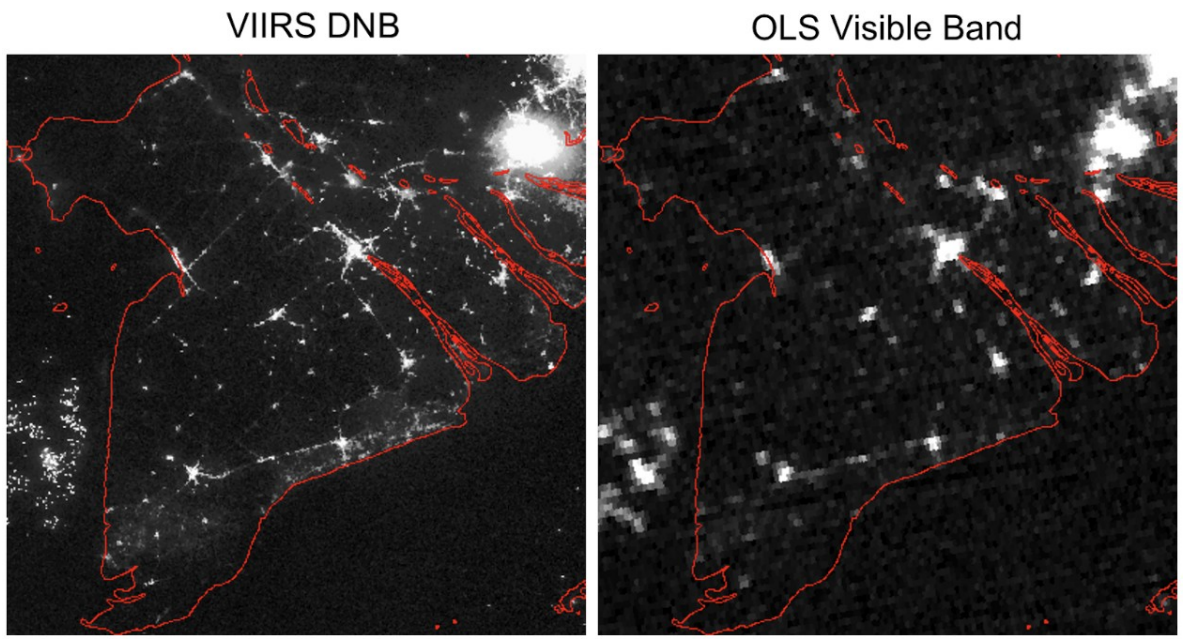
around 01:30 local time: same time everywhere

~500 m grid

15 arc-seconds; monthly and annual composites

Unit: $nW \cdot cm^{-2} \cdot sr^{-1}$ = nanowatt per cm^2 per steradian — upward-emitted radiance

Two sensor generations: DMSP-OLS and VIIRS



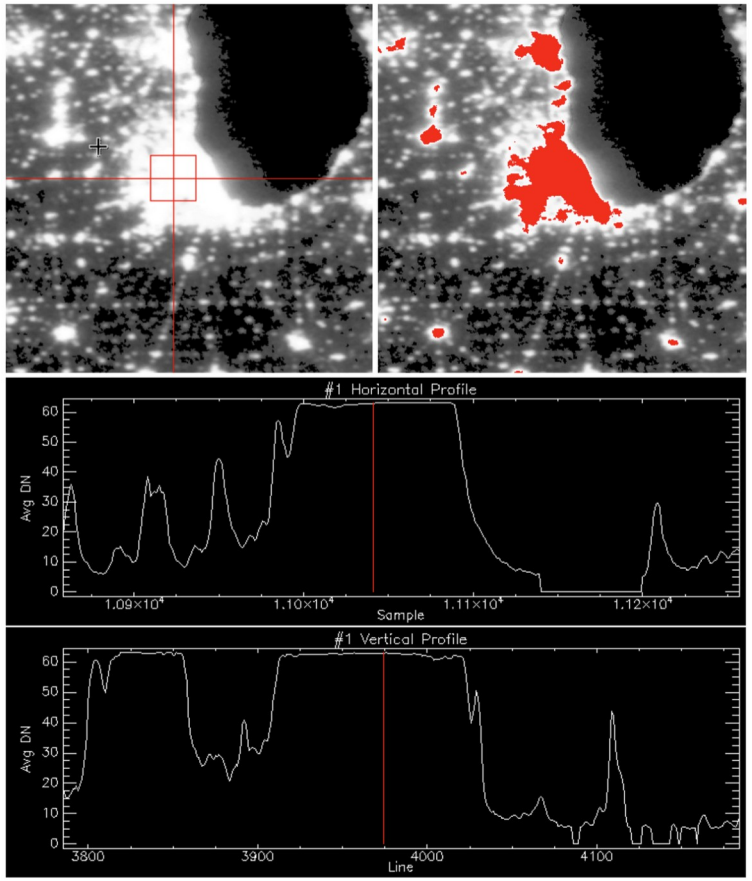
Mekong Delta, 23 Nov 2014: VIIRS DNB (left) vs DMSP-OLS (right). Source: Elvidge et al. (2013), via Open Nighttime Lights (CC BY 4.0).

Sensor break in 2013→2014: the notebook checks continuity at the junction (quality control 3.4)

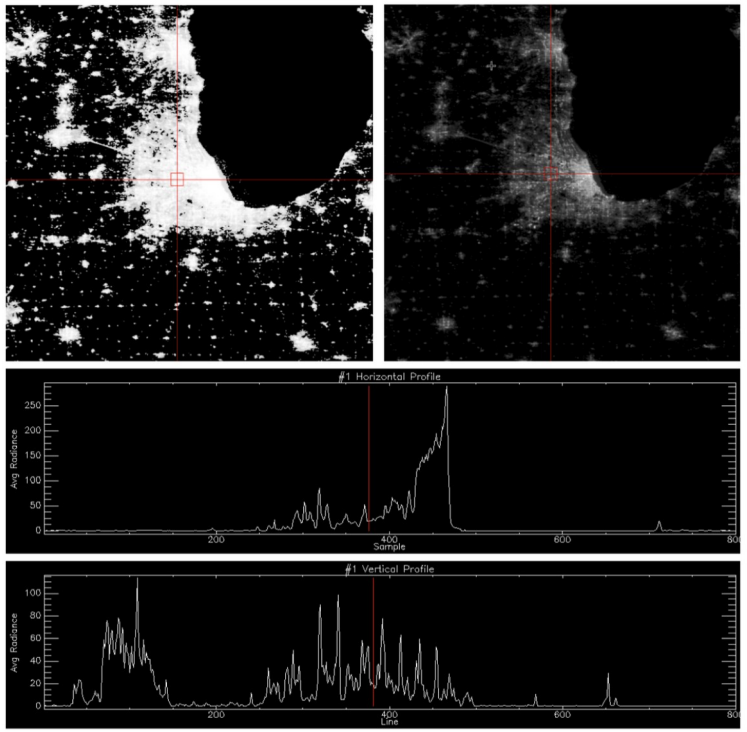
Sensor	DMSP-OLS	VIIRS DNB	Black Marble VNP46A2
Period	1992–2013	2012 →	2012 →
Resolution	~1 km	~500 m	~500 m, daily
Measurement	number 0–63, saturates	calibrated radiance	BRDF/moon-corrected radiance
Use here	long series (harmonised)	main analyses	shocks, day by day

Key point VIIRS is finer, calibrated and does not saturate. DMSP is only useful to go back before 2012, through a harmonised series to be read as a trend only.

Saturation and blooming: why VIIRS changes the game



DMSP-OLS 2013



VIIRS DNB 2016

Chicago: saturated pixels in red (DMSP). Profiles along the red lines. Source: Open Nighttime Lights (World Bank, CC BY 4.0).

Saturation

DMSP caps at 63: the city centre looks flat, its growth is invisible.

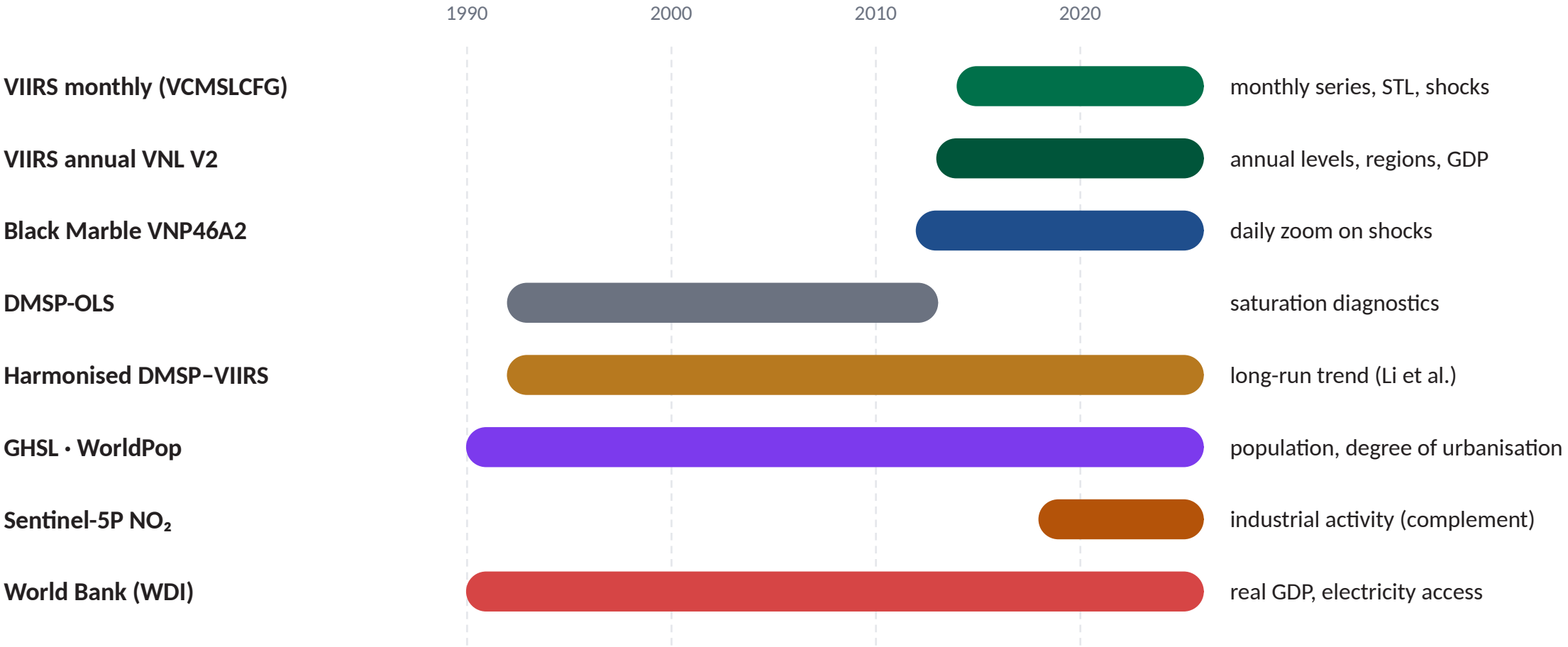
Blooming

Light "spills" around cities and inflates the lit area.

VIIRS: 25 to 280 nW

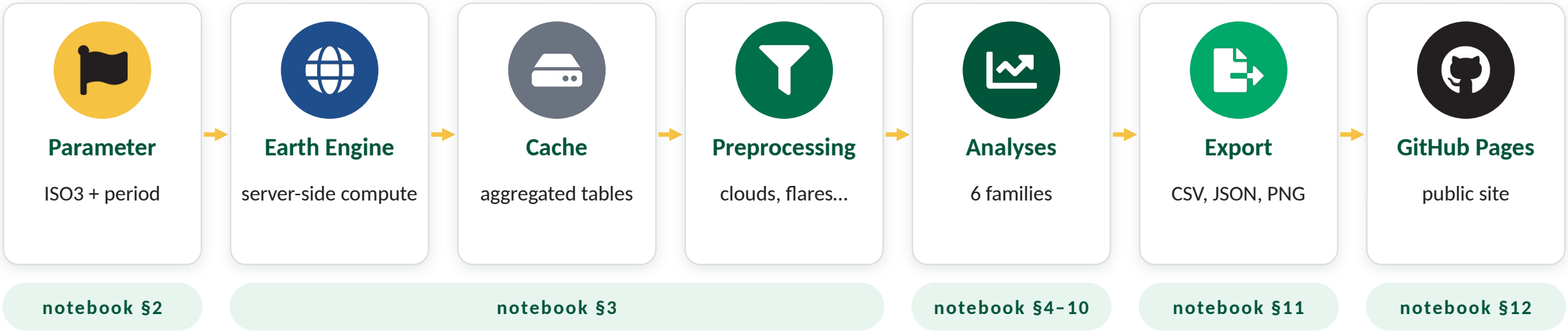
in the urban core that was capped

The pipeline's data catalogue



+ geoBoundaries administrative boundaries (FAO GAUL fallback) · ESA WorldCover 10 m · event catalogue (COVID-19, ACLED, EM-DAT)

The end-to-end pipeline: from one ISO3 code to a public site



Reproducible

no country-specific value is hard-coded: changing ISO3 is enough.

Lightweight

imagery stays in Earth Engine; only aggregated tables come back.

Cached

a second run does not query GEE again (FORCE_DOWNLOAD to force it).

Traceable

a manifest keeps parameters, versions and data availability.

Software architecture: one notebook, a src/ package, a static site

Acquisition

gee.py
connection, availability

boundaries.py
ADM0/1/2 boundaries

acquisition.py
zonal statistics

worldbank.py
WDI API

Preprocessing

preprocessing.py
clouds, flares, outliers

cache.py
caching

config.py
parameters, datasets

Analysis

timeseries.py
STL, anomalies, breaks

indicators.py
Theil–Sen, Gini, Theil, HHI

econometrics.py
elasticity, nowcast

shocks.py
events, impacts

pipeline.py
orchestration

Delivery

viz.py
charts, maps

messages.py
messages, confidence

export.py
static files

publish.py
GitHub Pages



Notebook (13 sections)

Colab · Kaggle · Jupyter; src/ code embedded and auto-unpacked



Dashboard (8 pages)

static HTML + JavaScript, data in dashboard/data/<ISO3>/

One cell to edit

```
# ⚙️  PARAMETERS — the only cell to edit
ISO3 = "TUN"
START_YEAR = 2014
END_YEAR = None         # last full year
ADMIN_LEVELS = ("ADM1", "ADM2")
GEE_PROJECT = None      # Google Cloud project
FORCE_DOWNLOAD = False # ignore the cache

CUSTOM_EVENTS = [       # country-specific shocks
  {"label": "Cyclone X", "start": "2019-03-01",
   "end": "2019-05-31", "type": "disaster"},
]

PUBLISH = False         # True -> GitHub Pages
REPO_NAME = "ntl-dashboard"
```

ISO3

3-letter country code (ISO 3166-1). TUN = Tunisia, SEN = Senegal, KEN = Kenya…

ADM1 / ADM2

1st and 2nd administrative levels: regions, then districts

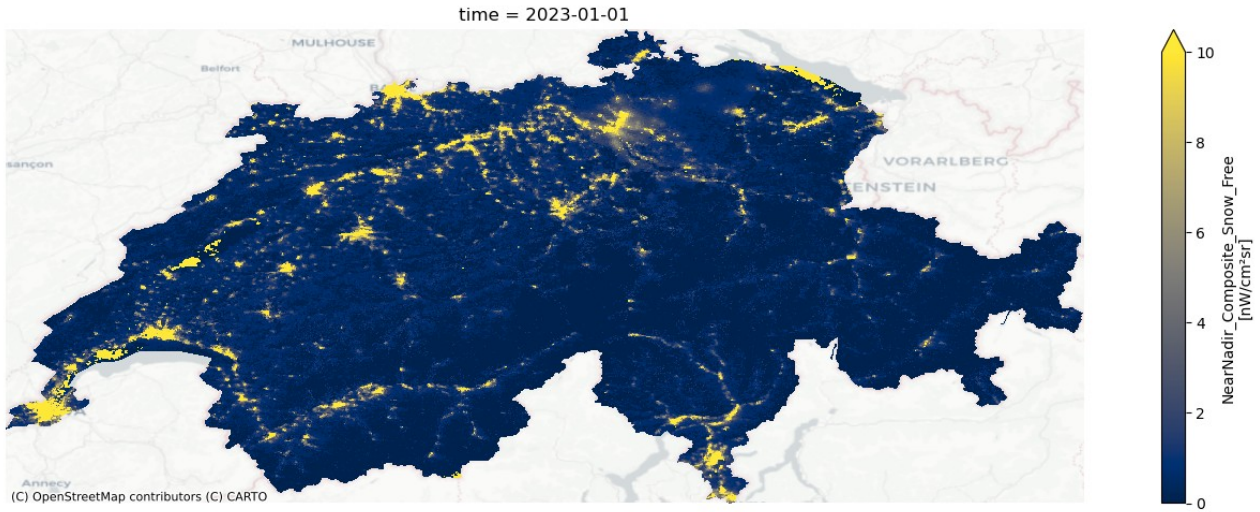
CUSTOM_EVENTS

your known shocks (cyclone, conflict, energy crisis) for section 8

Other thresholds

documented defaults in src/config.py, overridable in NTLConfig(…)

Basic quantities: radiance, SoL, light per capita



Monthly Black Marble (VNP46A3), Switzerland, January 2023 — real image. Source: blackmarblepy (World Bank, MIT).

Golden rule a pixel that was never observed (clouds) is not dark: it is missing. Treating it as zero manufactures a false recession.

Pitfall comparing the SoL of two regions of very different sizes: also reason per capita and as a national share.

Pixel radiance

$$r_i \ [\mathrm{nW \cdot cm^{-2} \cdot sr^{-1}}]$$

light intensity measured over ~500 m

Sum of lights (SoL)

$$\mathrm{SoL} = \sum r_i \quad (i \in \mathrm{unit})$$

an activity index, not an energy

Light per capita

$$\mathrm{SoL} \ / \ \mathrm{population}$$

separates an active region from a populous one

The 6-step preprocessing chain

1 Clouds

pixel valid if cf_cvg ≥ 2; unit-month kept if ≥ 80% of pixels are valid, then rescaled

Why: *don't mistake a cloudy month for a recession*

2 Negatives

values < 0 set to 0

Why: *calibration noise over dark areas*

3 Outliers

cap of 1,000 nW; spike > 5 × the pixel's annual median → median

Why: *fires, lightning, glints*

4 Background

pixels unlit in the VNL V2 annual composite → 0

Why: *EOG already masks background and ephemeral lights*

5 Gas flares

≥ 50 nW, < 5 inhabitants, not built-up (union of years) + 2 km halo

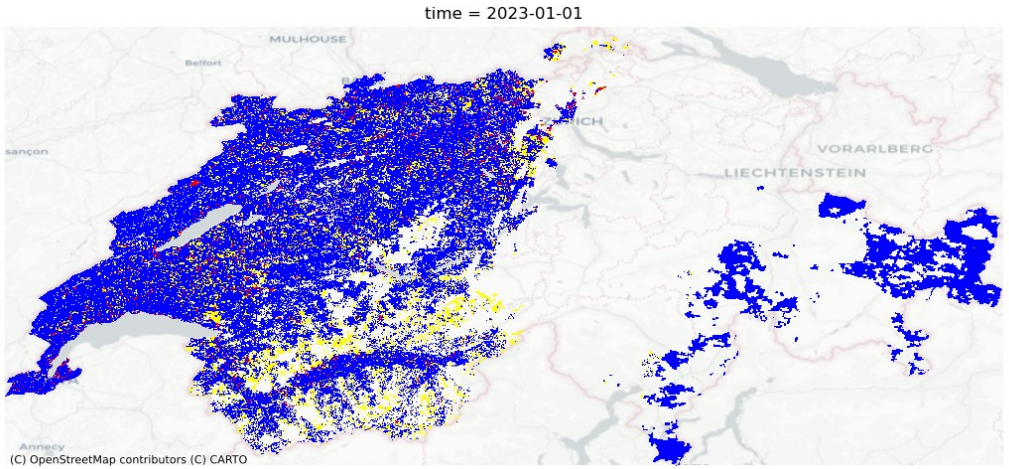
Why: *removes flared gas and its spill-over*

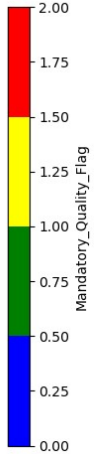
6 DMSP

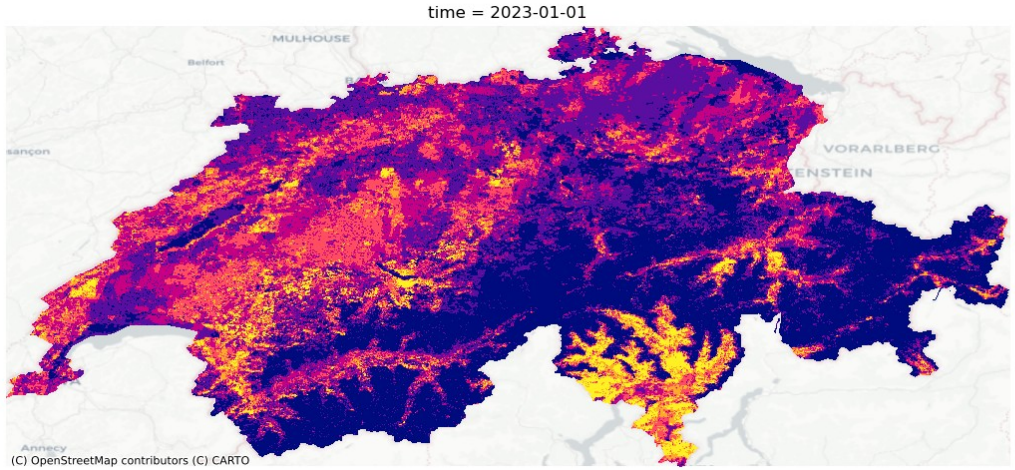
share of saturated pixels measured; DN > 7 threshold on the harmonised series

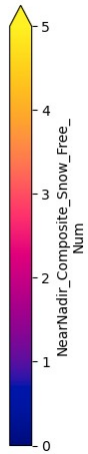
Why: *recommendation of Li et al. (2020)*

Clouds and quality: a missing pixel is not a dark pixel









Daily quality flag (VNP46A2), Switzerland, 1 Jan 2023

Number of snow-free observations (VNP46A3), January 2023

Real images — Source: NASA Black Marble via blackmarblepy, "quality-assessment" notebook (World Bank, MIT).

Quality flags

Daily Black Marble: 0 = high-quality persistent, 1 = high-quality ephemeral, 2 = poor quality (clouds, moon…). The notebook keeps high quality only, cloud- and snow-free.

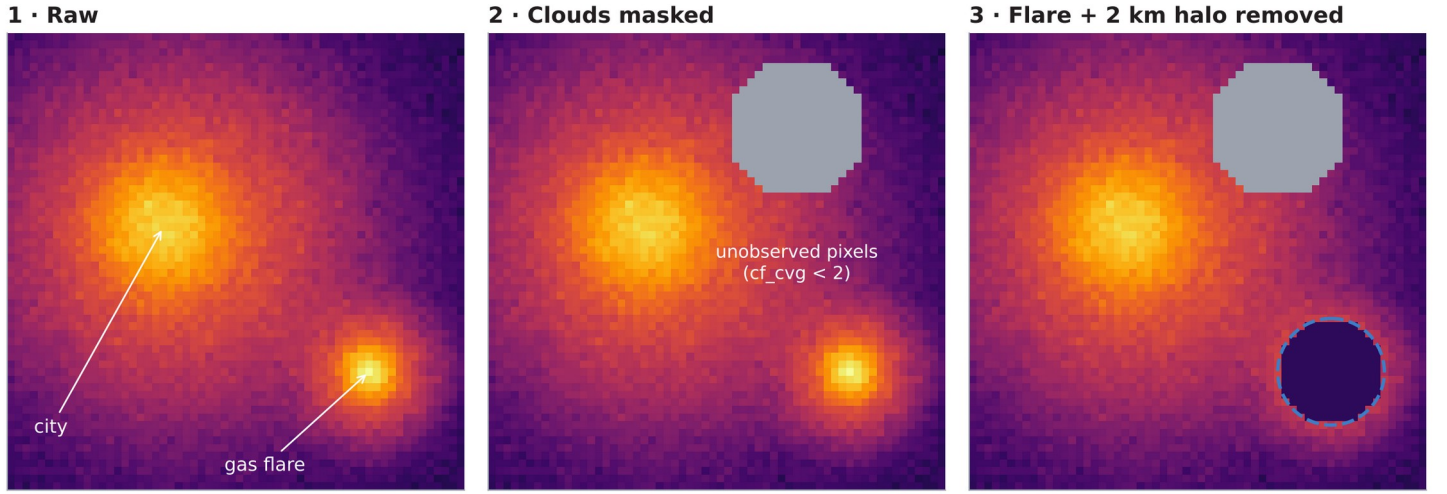
Coverage cf_cvg

Monthly VIIRS: number of cloud-free observations per pixel. In cloudy equatorial areas, one month may not be enough.

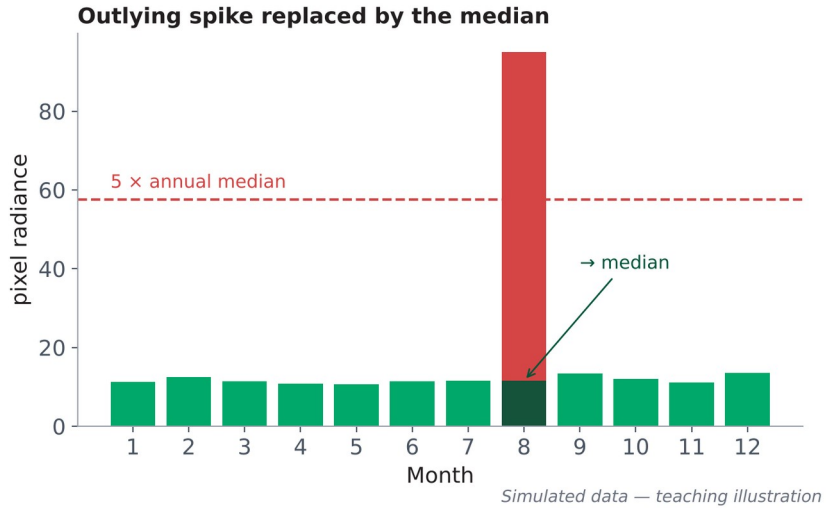
Rescaling

if 80–100% of pixels are valid, SoL is scaled up to the whole unit; below that, the month is excluded.

Gas flares and outliers: light without local activity



Simulated data — teaching illustration



Simulated data — teaching illustration

Detect

very bright (≥ 50 nW), nearly uninhabited (< 5 inhabitants), not built-up pixels, over the union of years

Vectorise

candidates become polygons, cached and exported for visual checking

Mask

the polygon + a 2 km halo is removed from every year

How to read the quality-control table shows the share of light removed as gas flares. In an oil-producing country it can be large: check the exported list (flare_candidates.csv) on a map.

Quality control: six indicators to read before any analysis

Months observed (national)

e.g. 132 / 132

gaps = broken series

Mean cloud-free coverage

e.g. 94%

< 80%: caution in the rainy season

Unit-months excluded

e.g. 3.1%

high: fragile regional series

Light removed (flares)

e.g. 0.8%

high: check the masked sites

Population source

GHSL — 2025

GHSL by default, WorldPop as an option

DMSP saturation

e.g. 0.45%

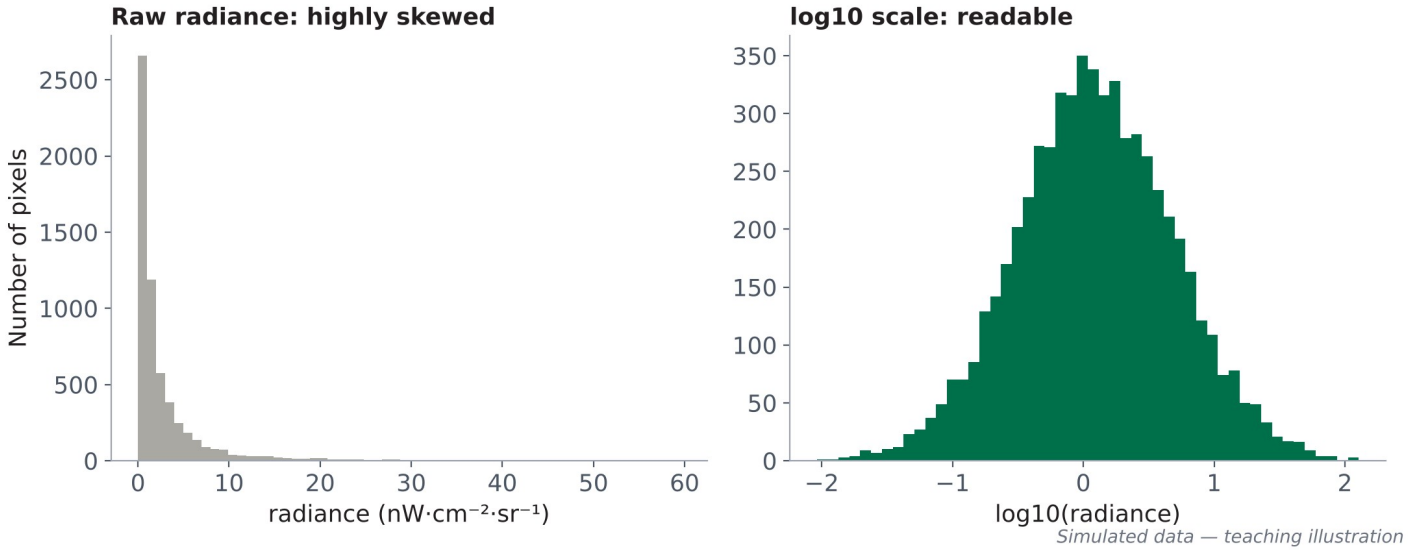
high: long series less reliable in cities

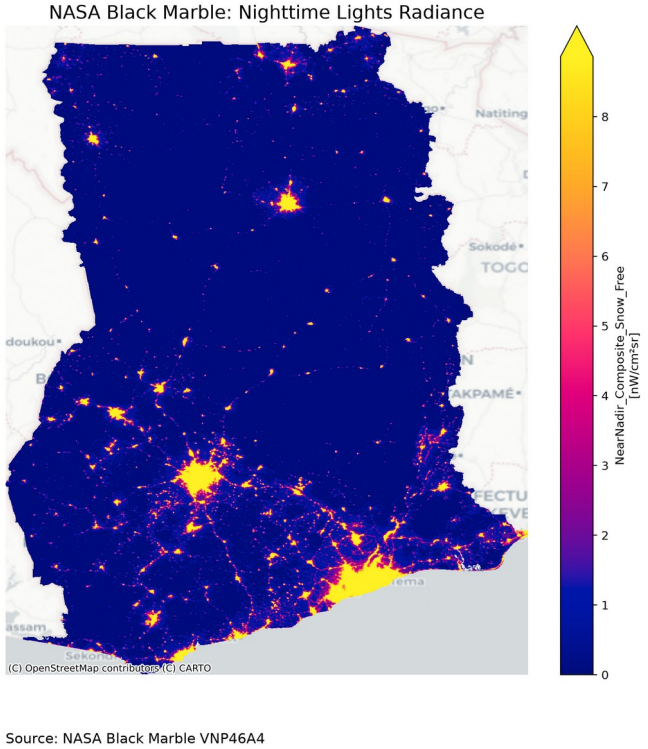
DMSP → VIIRS junction

The notebook compares growth 2012→13, the 2013→14 junction and 2014→15. A jump at the junction very different from neighbouring years signals a residual sensor break: the long series must then be read with caution.

How to read the “e.g.” values are format examples. If any indicator is high, the results that depend on it are less reliable.

Exploration: look before you compute





Annual Black Marble 2025, Ghana (real). Source: blackmarblepy (MIT).

Interactive map

annual composite with a time slider

Before / after

first vs last year, split map

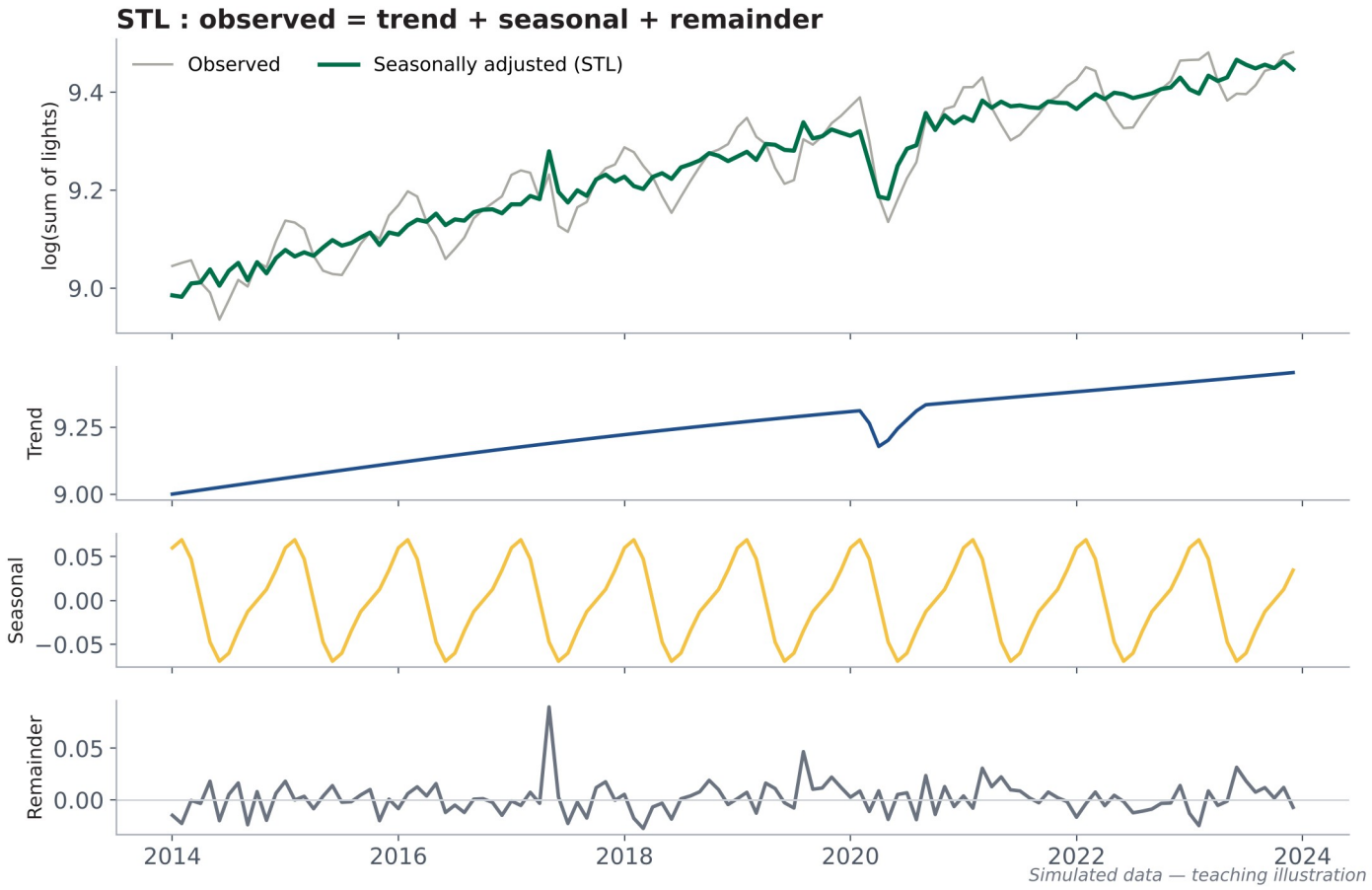
ADM1 statistics

SoL, national share, light per capita

How to read radiance is highly skewed: a few very bright urban pixels, many faint ones. Hence mapping on a log scale.

Pitfall mistaking a residual flare or a very bright isolated site for a city

Seasonality and STL decomposition



Economic question

Is light activity really rising once seasonal effects are removed?

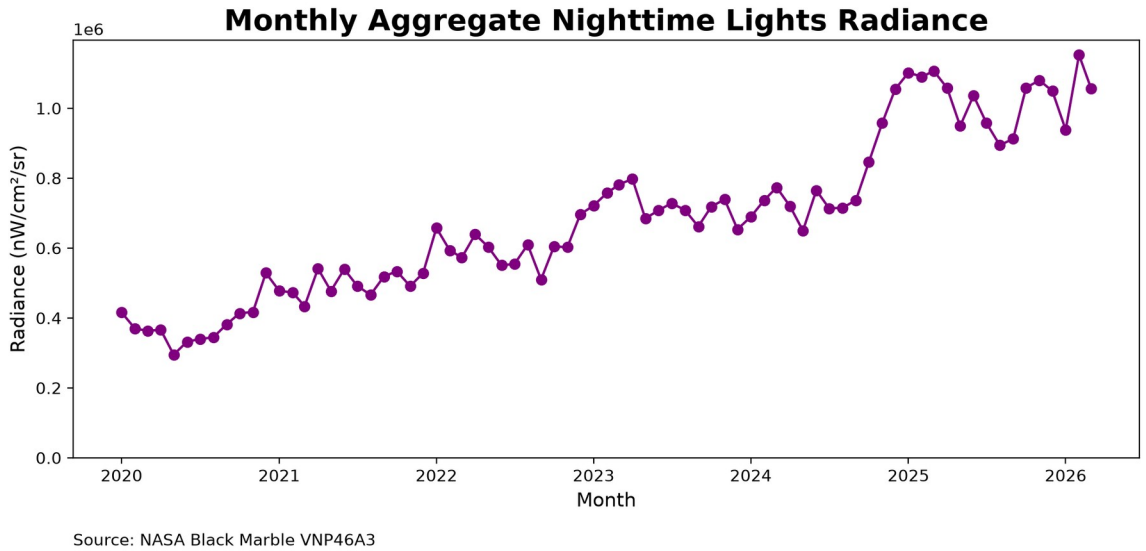
$$\log(\mathrm{SoL}_t) = T_t + S_t + R_t$$

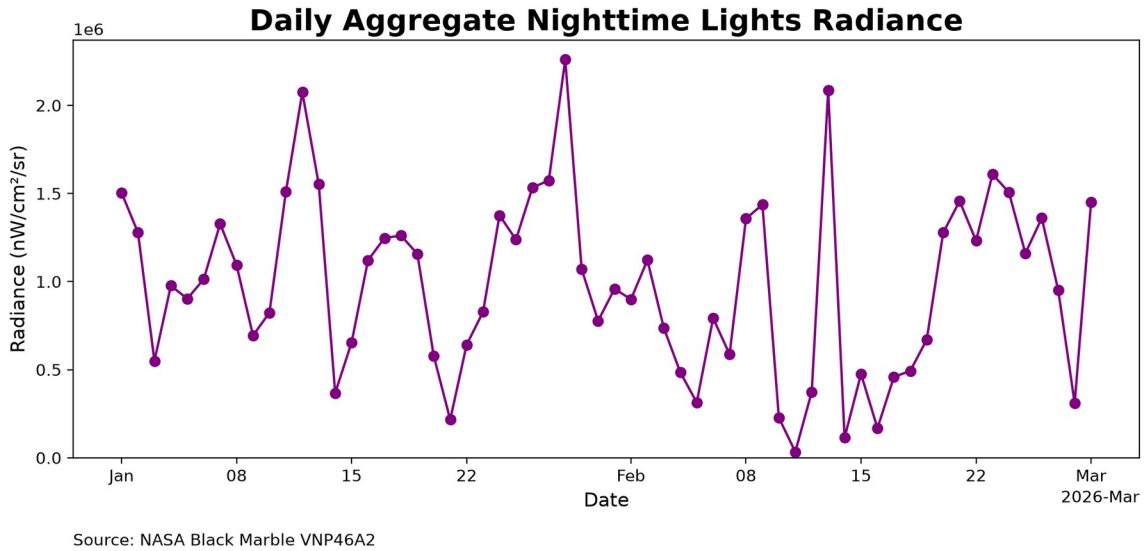
STL = Seasonal-Trend decomposition using Loess. Applied to the log, period 12, robust version that resists atypical months.

How to read read the seasonally adjusted curve (green). Year-on-year change (same month, previous year) also neutralises the season.

Pitfall comparing December with November: festivities, snow and night length distort the reading.

On real data: monthly or daily?





Monthly (VNP46A3), Ghana, 2020–2026: readable trend

Daily (VNP46A2), Ghana, Jan–Mar 2026: very noisy

Real images — Source: NASA Black Marble via blackmarblepy (World Bank, MIT).

Monthly → trends

STL, year-on-year change, regional growth, elasticity

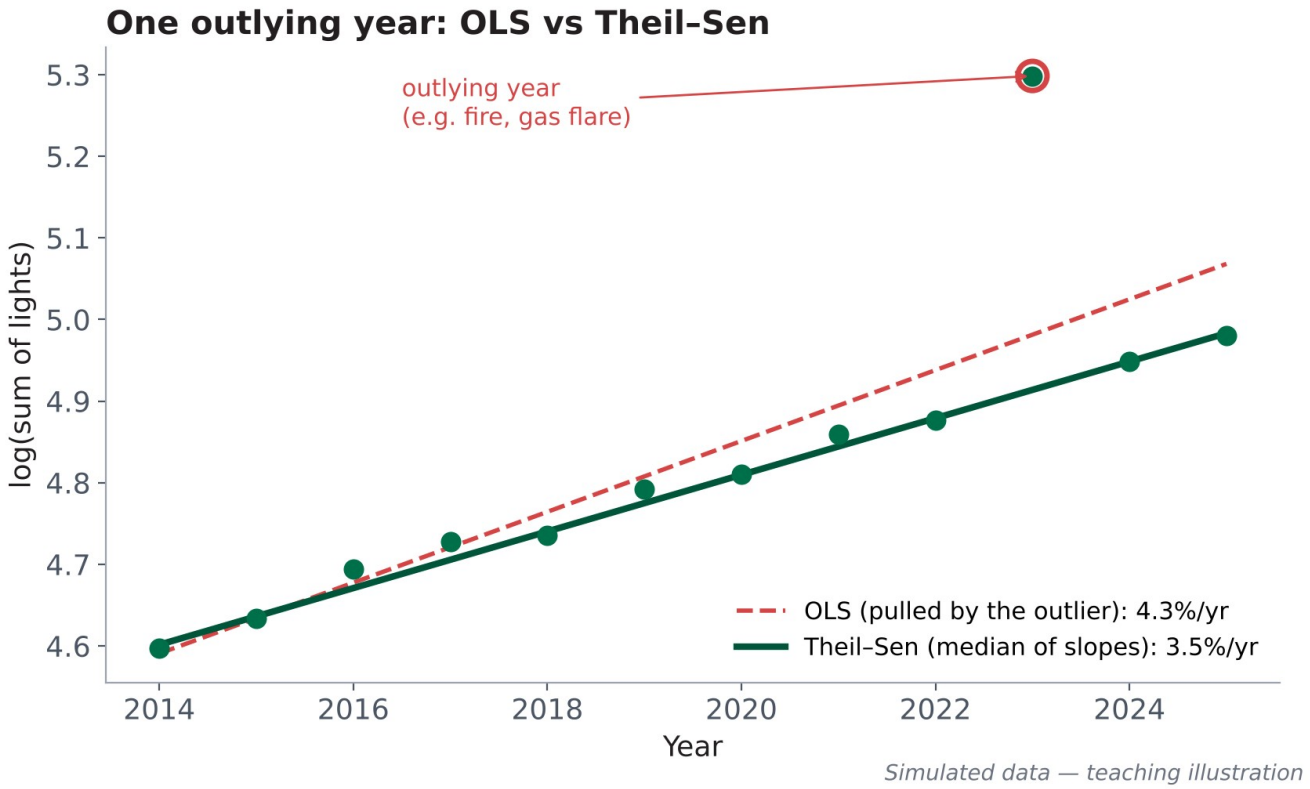
Daily → dated shocks

base-100 index + 7-day rolling median, high quality only

Annual → levels

masked VNL V2 composites: regional levels, GDP, inequality

Measuring a robust trend: Theil–Sen and Mann–Kendall



Theil–Sen slope

median of the slopes between all pairs of years, on log(SoL). An outlying year barely moves it.

$$\beta = \text{median}\ [\ (y_j - y_i)\ /\ (t_j - t_i)\ ]$$

Mann–Kendall test

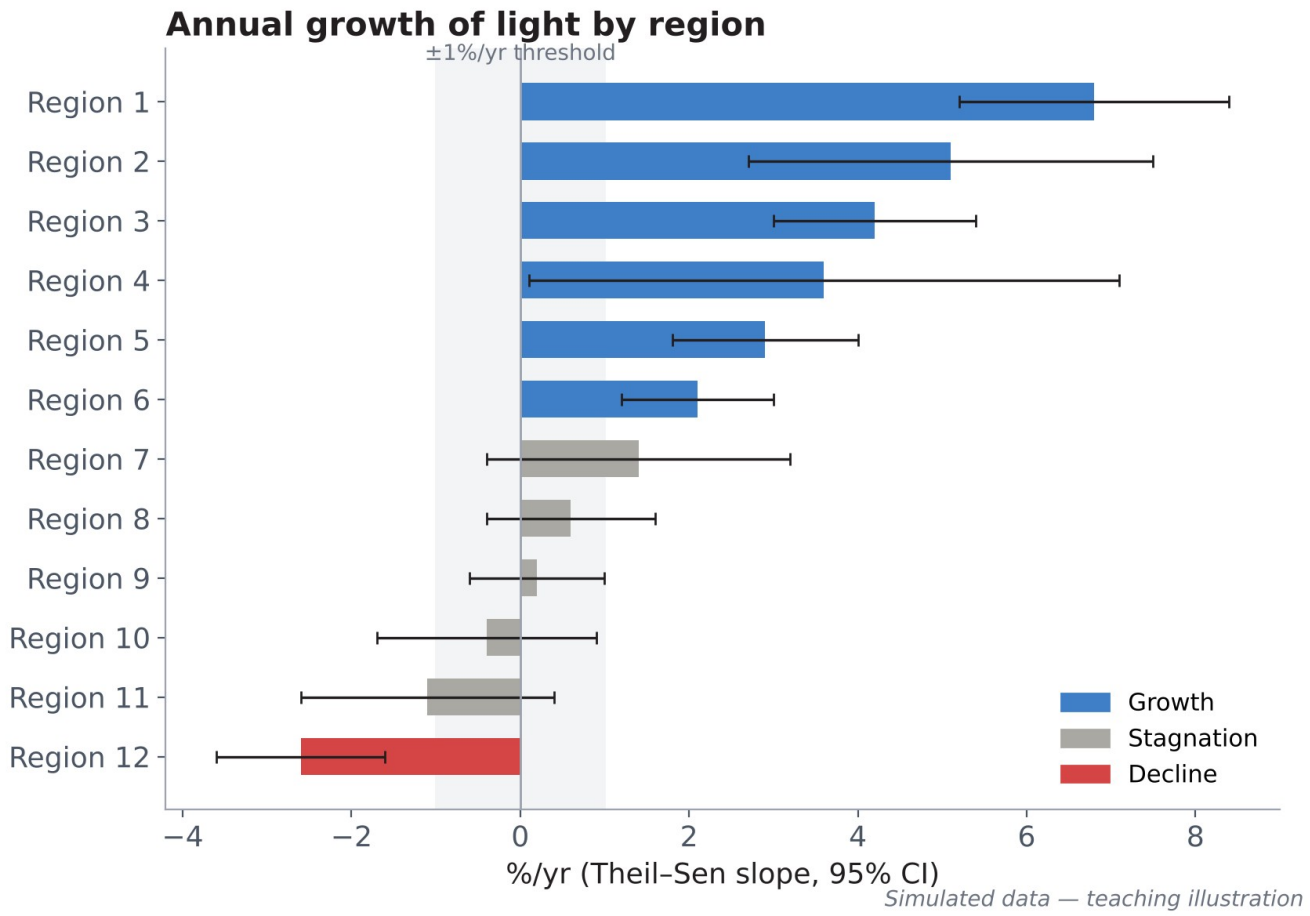
is the trend significant or due to chance? Counts rising vs falling pairs.

$$S = \Sigma\ \text{sign}(y_j - y_i)$$

How to read annual growth = exp(β̂) − 1, with its 95% confidence interval.

Class “Growth” or “Decline” if |slope| ≥ 1%/yr AND p < 0.10; otherwise “Stagnation”

Who grows, who stagnates, who declines?



Economic question

Where is activity rising, and where is it falling, over the period?

Growth — slope ≥ +1%/yr and significant

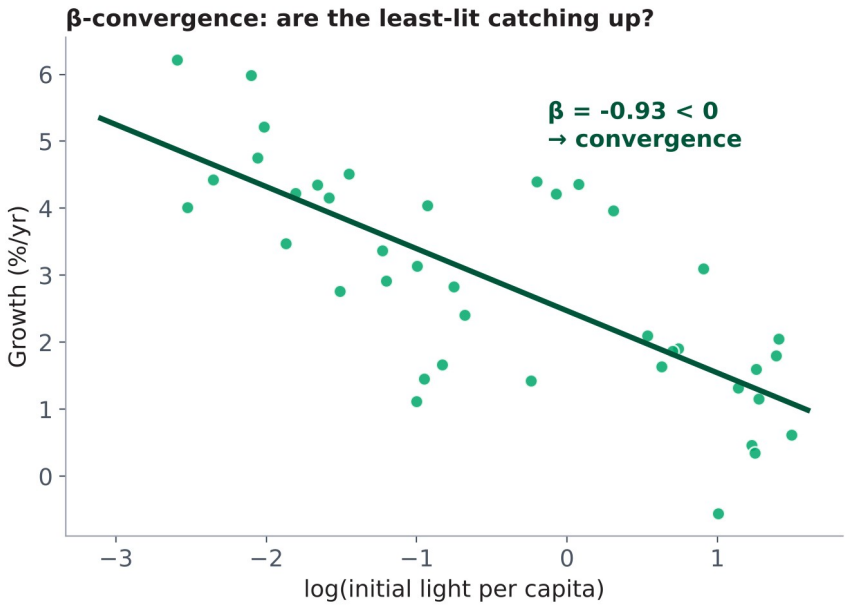
Stagnation — |slope| < 1%/yr, or trend not established

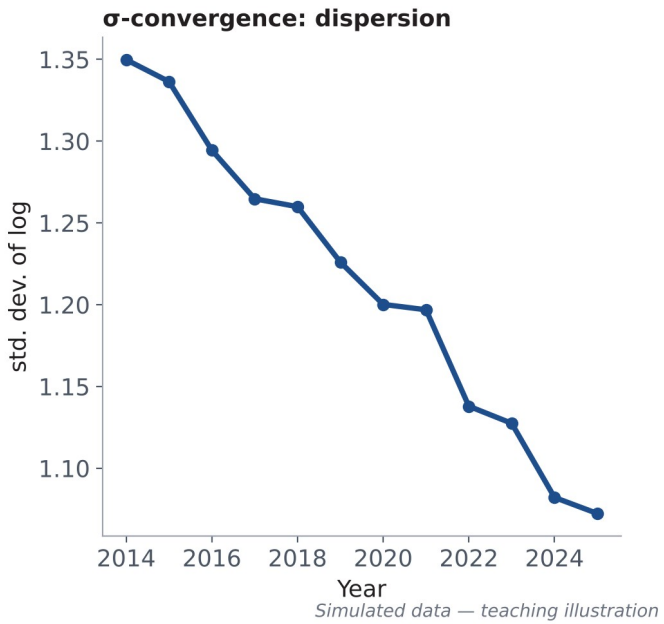
Decline — slope ≤ −1%/yr and significant

How to read error bars are 95% CIs. Region 7 has a positive slope but its CI touches zero: it stays in "stagnation".

Pitfall reading "stagnation" as "standing still": it often means "not enough information".

Convergence and concentration: is activity spreading out?





Economic question

Does growth benefit peripheral regions, or concentrate around the capital?

NO_2 (Sentinel-5P)

Spearman correlation with light: a complementary industrial indicator.

β-convergence

$$g_i = a + \beta \cdot \log(x_{i,0}) + \varepsilon_i$$

β < 0: initially dimly lit regions grow faster — they catch up.

σ-convergence

$$\sigma_t = \text{std. dev. } [\log(x_{i,t})]$$

σ falls: gaps between regions narrow.

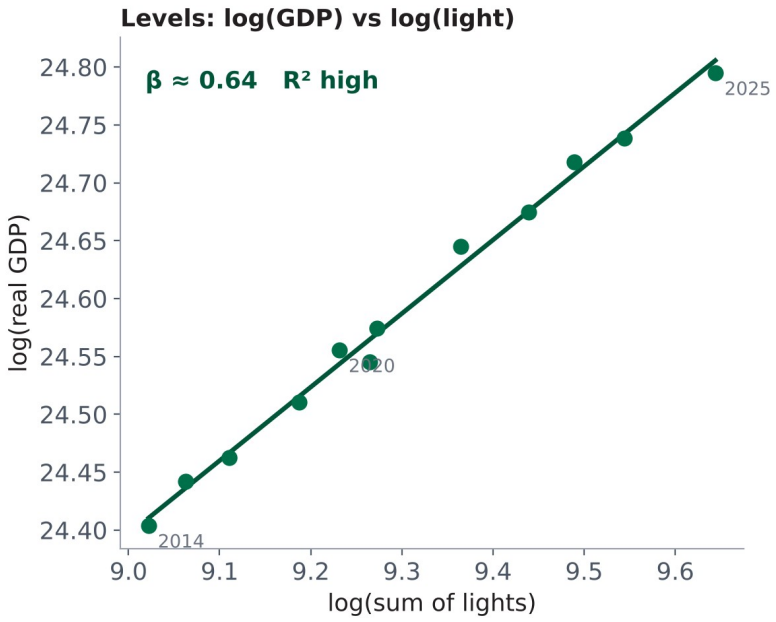
Herfindahl (HHI)

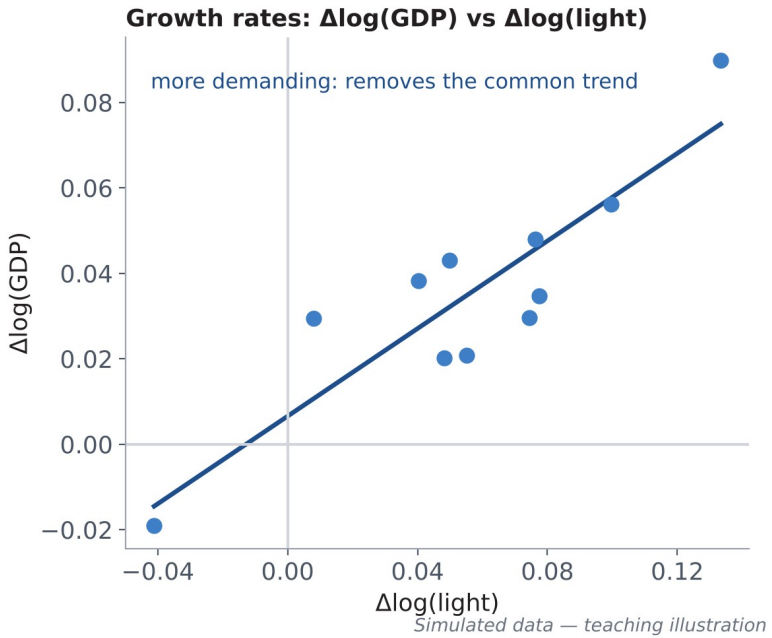
$$HHI = \Sigma s_i^2$$

s_i = share of national light; 1 = everything in one unit.

Economic reading rising concentration = polarisation (capital, ports); convergence = electrification spreading to the periphery.

The lights–GDP relationship: estimating an elasticity





$$\log(NTL_t) = \alpha + \beta \cdot \log(GDP_t) + \varepsilon_t$$

β = elasticity: +1% GDP ↔ +β% light.

Levels (VIIRS) — long-run relationship; spurious-regression risk

Growth (VIIRS) — Δlog: removes the common trend; more demanding

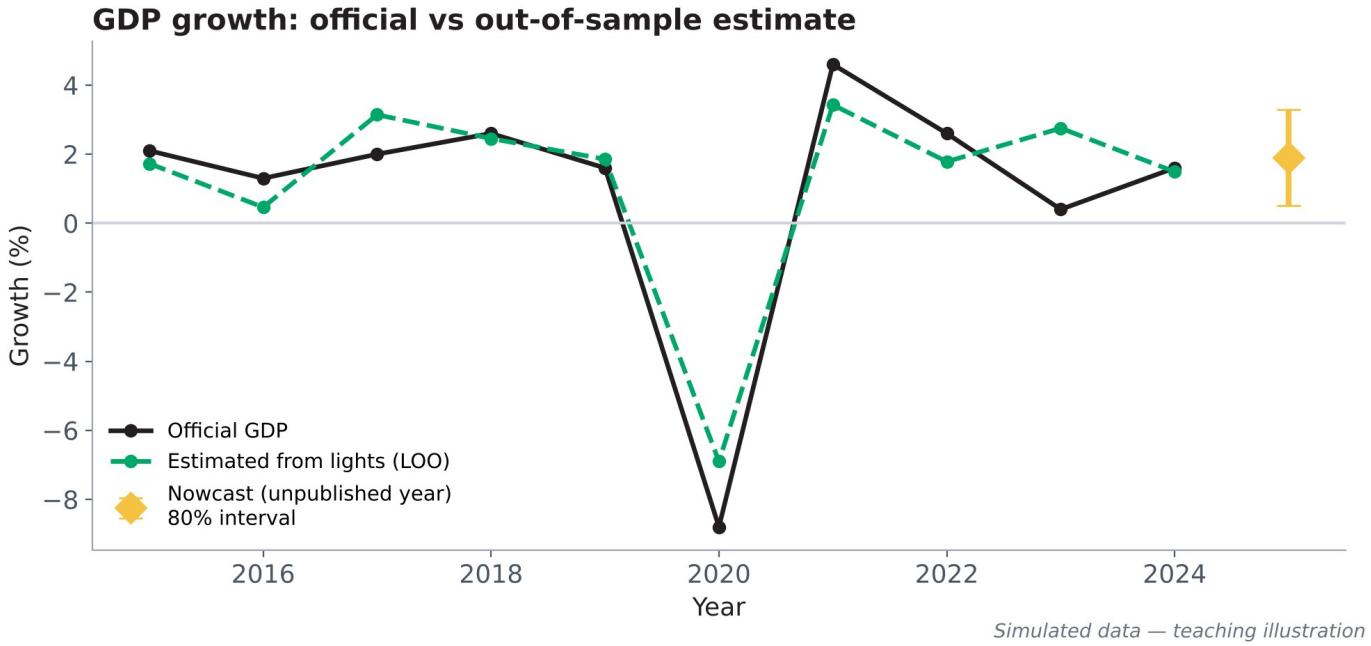
Long series — harmonised DMSP–VIIRS, with a sensor dummy

Why two specifications?

Two series that both rise over time look correlated even with no causal link: this is spurious regression. If β stays similar in growth rates, the relationship is credible.

Economic reading β < 1: light grows more slowly than GDP (services, LEDs, efficiency). β unstable across specifications: use lights only as a directional signal.

Diagnostics and out-of-sample prediction: an honest test



HAC (Newey–West)

standard errors corrected for autocorrelation in annual residuals

Durbin–Watson

residual autocorrelation; ≈ 2 if absent

Breusch–Pagan

heteroskedasticity: does error variance change?

Jarque–Bera

normality of residuals

Out-of-sample prediction (leave-one-out)

Each year is predicted by a model estimated without it. The notebook reports the correlation and the mean absolute error (in growth points): the honest measure of predictive ability.

Pitfall settling for the R^2 in levels: it is almost always high. Read β's CI and the out-of-sample error.

Estimating activity where statistics are missing

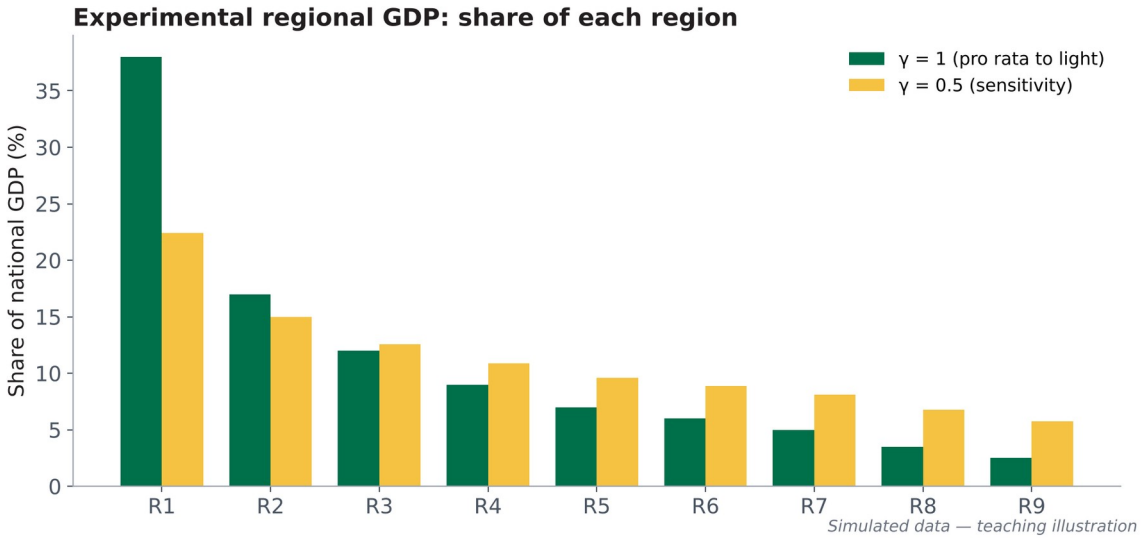
Nowcast: the not-yet-published year

GDP growth for the recent year is estimated from light growth, with 80% and 95% prediction intervals.

$$\Delta\log(GDP_t) = a + \gamma \cdot \Delta\log(NTL_t)$$

Status: experimental statistics

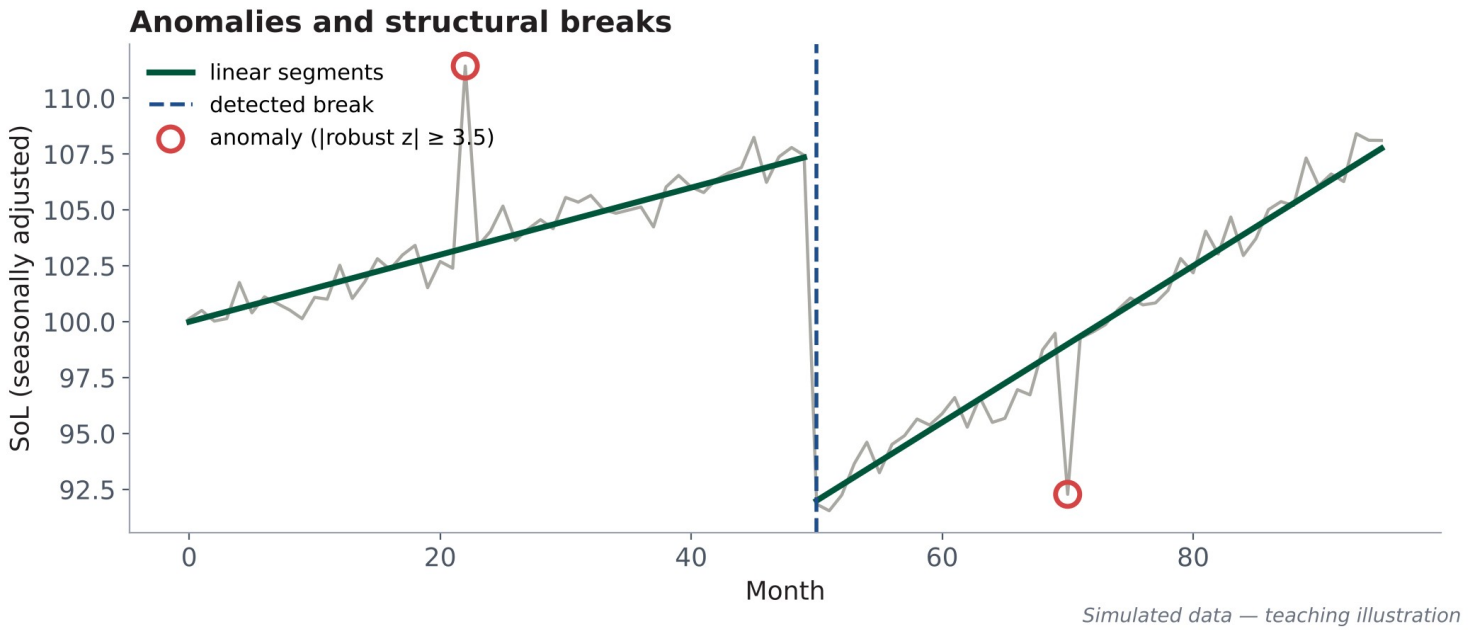
These estimates carry “low” confidence by construction: extrapolation from a model estimated on few years. Publish with intervals and a methodological note, after comparison with available regional accounts.



$$GDP_r = \text{national GDP} \times SoL_r^{\gamma} \,/\, \Sigma\, SoL^{\gamma}$$

Experimental regional GDP. γ = 1: pro rata to light; γ = 0.5: sensitivity giving more weight to dimly lit regions. The gap between the two measures the uncertainty due to the assumption.

Detecting shocks: anomalies and structural breaks



Simulated data — teaching illustration

Anomaly

month whose STL remainder is extreme: |robust z| ≥ 3.5 AND deviation ≥ 5%

$$z = (R_t - \text{median}) \,/\, (1.4826 \times \text{MAD})$$

Structural break

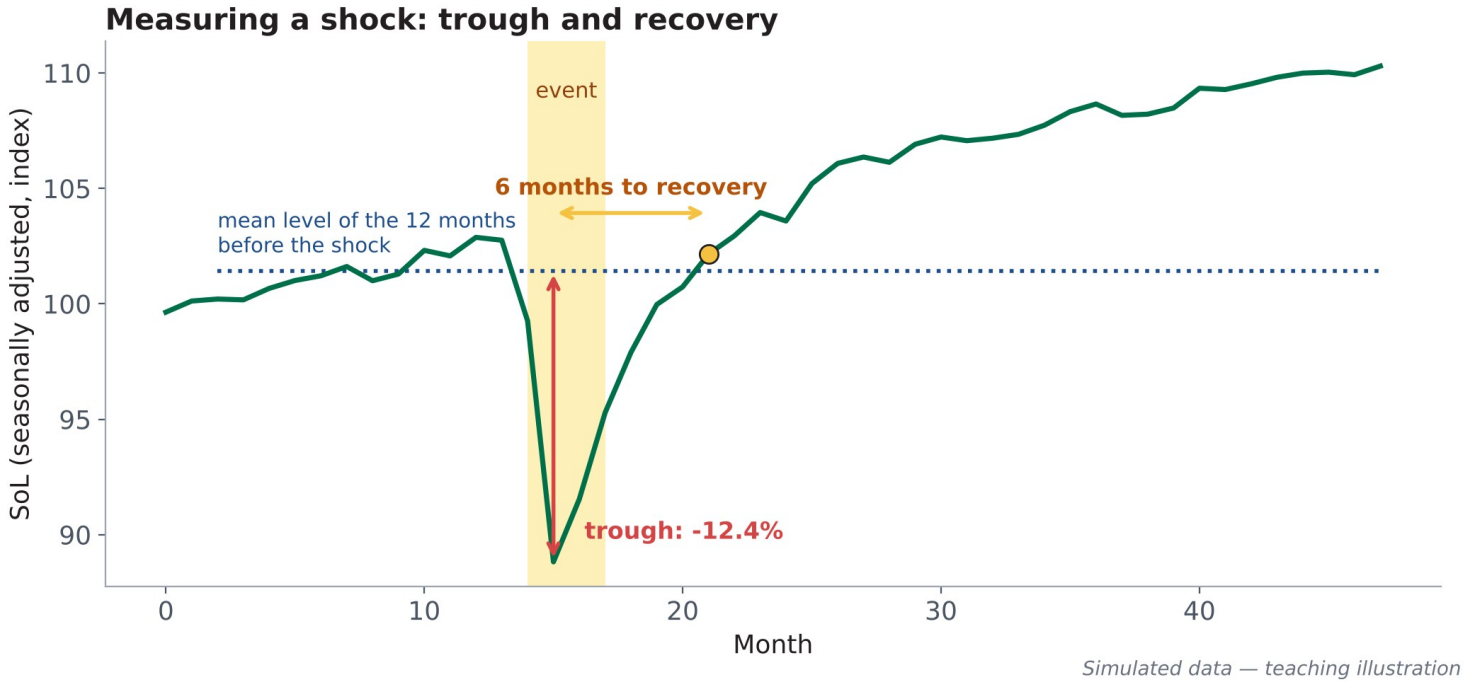
lasting change in level or slope; piecewise linear segmentation, BIC-type penalty, ≥ 12 months on each side

MAD = median absolute deviation: a dispersion measure robust to extreme values.

Event catalogue

COVID-19 (generic, March 2020 – June 2021) + your CUSTOM_EVENTS + optional ACLED (conflict) or EM-DAT (disaster) export. A break with no known event within ±6 months is flagged: it must be documented (sensor, outage) before being interpreted.

Measuring a shock: depth of the trough and speed of recovery



Pre-shock level

mean of the 12 months before the event, seasonally adjusted series

Trough

maximum deviation from that level during the event (in %)

Recovery

number of months to return to the pre-shock level

Distinguishable from noise?

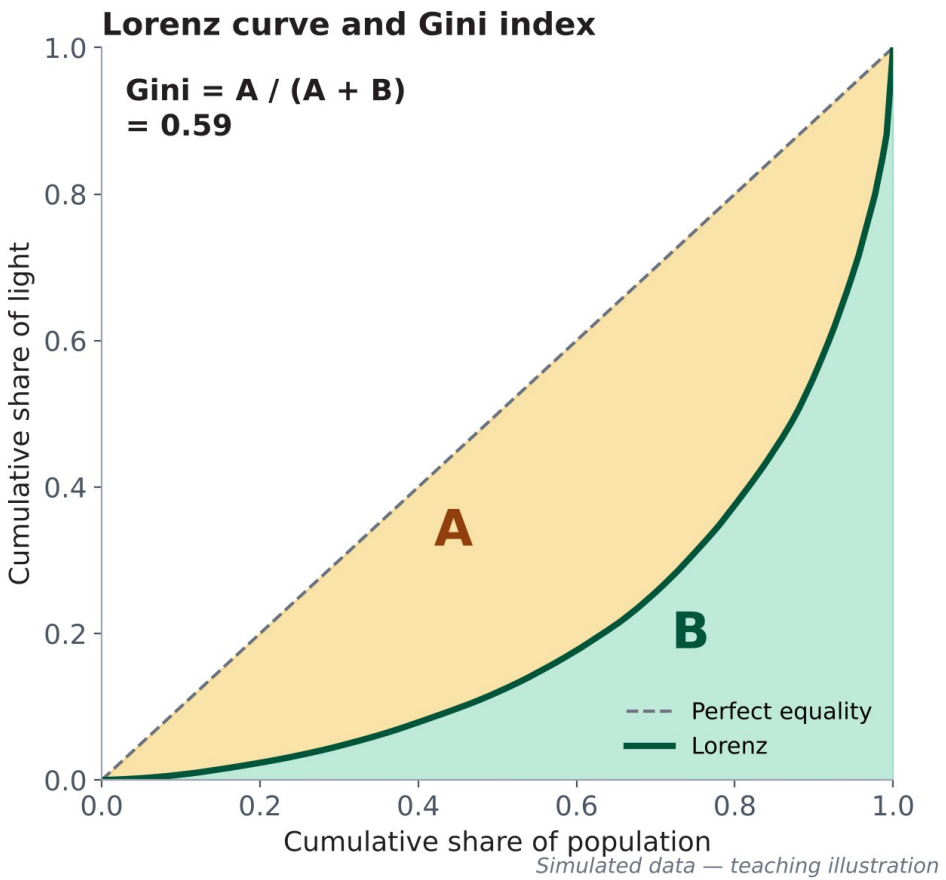
the trough is compared with pre-shock monthly variability

Daily Black Marble zoom

To date a shock to the day: base-100 index (median of the first 60 days), high-quality cloud- and snow-free pixels, 7-day rolling median. Impact also computed by region (heat table).

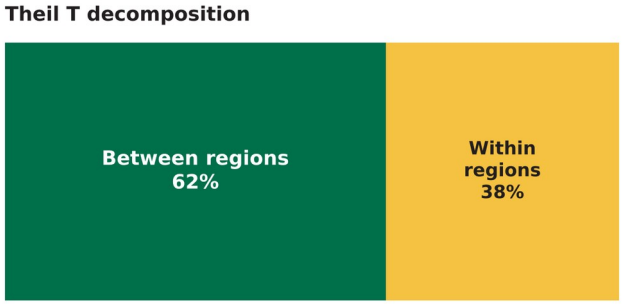
Reading light captures stoppages of night-time activity well (lockdowns, outages, displacement); it underestimates agricultural or daytime-service shocks.

Spatial inequality: Lorenz, Gini and Theil



Simulated data — teaching illustration

How to read curve far from the diagonal = high inequality. Gini = area A / (A + B).



Simulated data — teaching illustration

Economic question

Is access to light, a proxy for development, fairly distributed across people?

Gini

0 = equality, 1 = all light in a single unit. Population-weighted, on the finest units.

Theil T

decomposes into between-region and within-region inequality: at which level to act?

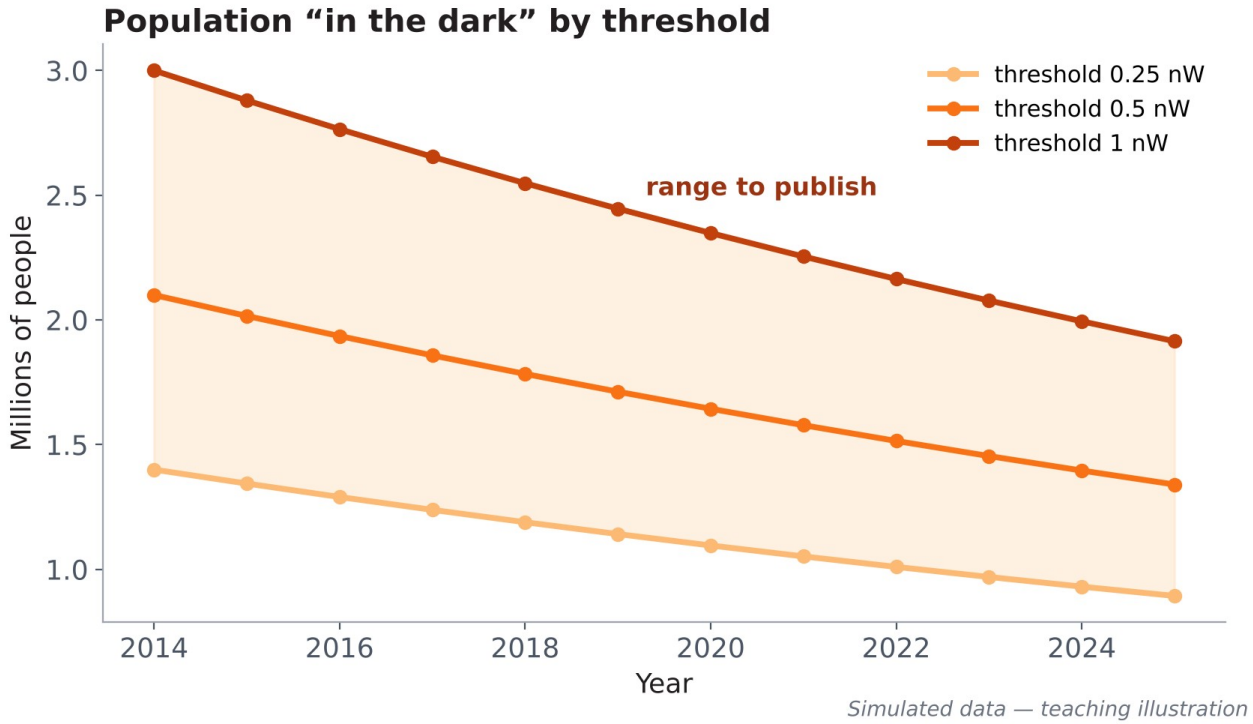
Change in Gini

95% bootstrap interval (300 draws): is the rise or fall significant?

Urban / rural gap

urban ÷ rural light per capita (GHSL degree of urbanisation)

The population “in the dark”: a proxy for electricity access



Definition

Inhabitants of populated pixels (≥ 5 inhabitants, ~0.21 km²) whose radiance stays below the detection threshold (0.5 nW by default).

thresholds tested: $0.25 \cdot 0.5 \cdot 1$ nW

World Bank comparison

gap with the EG.ELC.ACCS.ZS access rate (household surveys)

Pitfall “inhabited without light” ≠ “not connected”: a connected low-consumption household is invisible; the result depends on the threshold and the population map.

Publishing always publish a range (by threshold), never a figure to the nearest person. The map of dark areas helps target electrification.

The confidence system: high, medium, low

Example: NTL–GDP elasticity (src/econometrics.py)

Rules tested in order; the first true one wins

1. **n < 8 years** → LOW
   no
2. **p < 0.01 · CI < 1 · R^2 ≥ 0.6 · n ≥ 15** → HIGH
   no
3. **p < 0.01 · CI < 1 · R^2 ≥ 0.6 (n < 15)** → MEDIUM
   no
4. **p < 0.10** → MEDIUM
   no
5. **otherwise** → LOW

High

significant, sufficient sample, insensitive to choices → publishable message

Medium

significant but short sample or threshold-sensitive → publish with caveats

Low

not significant, extrapolation, strong dependence on assumptions → a lead, not a conclusion

Trends follow a similar rule (n ≥ 6; p < 0.01 → high; p < 0.10 → medium). Every message shows its confidence AND its reason.

Publishing light-based statistics: five good practices

Always an interval

95% CI, range by threshold, prediction interval: never a bare number.

Experimental status

Regional GDP and nowcast: labelled “experimental statistics”.

Traceability

manifest: parameters, versions, data availability, extraction date.

Documented thresholds

every threshold is shown on the dashboard's “Methods & limitations” page.

Never alone

as a complement to accounts and surveys, never a substitute.

Recommendations produced automatically by the notebook (section 10)

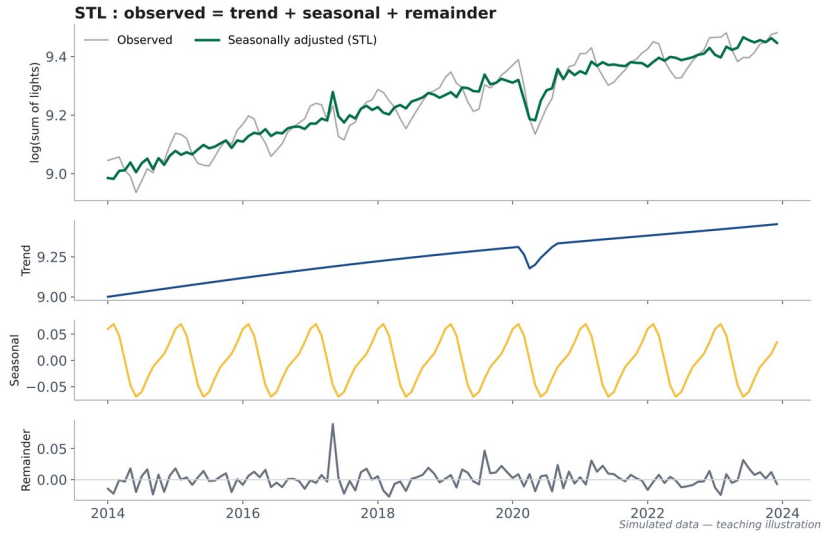
Use lights as a leading, subnational indicator · Do not release GDP estimates if the NTL–GDP link is weak · Cross the “in the dark” map with connection registers · Document unexplained breaks · Update every quarter.

The dashboard: 8 pages, no server

https://<user>.github.io/ntl-dashboard/?country=TUN

Overview | Map | Trends | Regions | Shocks | Inequality | Methods | Data

+3.5%/yr	0.62	0.54	1.4–3.0M
growth of light	light–GDP elasticity	light per capita Gini	people in the dark



Key messages

- 🟢 National trend
- 🟠 Regions
- 🟠 Link with GDP
- 🔴 Recent estimate
- 🟠 Inequality and access

Illustrative mock-up (dummy values)

Static export (section 11)

simplified GeoJSON, CSV, JSON, PNG under dashboard/data/<ISO3>/ — GEE tiles expire, a PNG does not.

Multi-country

countries.json: several countries can coexist on the same site.

Manifest

parameters, versions, references: the reproducibility guarantee.

Publish in one call, with a secure token

publish_dashboard(ISO3, REPO_NAME, GITHUB_USER)



Check the token

GET /user



Create the public repo

if it does not exist



Push the site

gh-pages branch, one timestamped commit



Enable Pages

and wait for the build



Show the URL

<user>.github.io/<repo>



Token security: read before running

Fine-grained token, short expiry (7–30 days), limited to the dashboard repository only.
Minimum permissions: Contents and Pages (read/write); Administration only to create the repository.
Storage: Colab secret GITHUB_TOKEN or .env file — never in code, outputs or URLs.
Before sharing: clear the publishing cell's output, then revoke the token.

Limitations and biases: what must always be said

What lights measure

radiance at ~01:30; agriculture, daytime trade and rural informal economy poorly captured

Saturation, blooming

DMSP saturates; blooming still inflates the lit area

Flares, isolated sites

the mask may remove a mine or miss a small flare

Clouds, seasons, moon

partially observed months; STL removes the average season, not an abnormal one

Product breaks

DMSP and VIIRS not comparable; modelled long series, trend only

NTL–GDP elasticity

10 to 30 points: wide intervals, non-causal and unstable relationship

Electricity access

“inhabited without light” ≠ “not connected”; threshold-sensitive

Administrative boundaries

geoBoundaries / GAUL may differ from official boundaries

What lights have made it possible to measure

The activity trend

seasonally adjusted annual growth, with CI

The geography of growth

regions growing, stagnating, declining

Convergence

catching up or polarisation

A GDP signal

elasticity, nowcast, experimental regional GDP

The impact of shocks

trough, recovery, day-level dating

Inequality and access

Gini, Theil, population in the dark (range)

One pipeline, one parameter, and for every result: a value, an interval, a confidence level and its reason.

Towards your own projects

Sub-annual tracking

a quarterly leading indicator of regional activity for your country

Assess a shock

cyclone, conflict, energy crisis: trough, recovery, affected regions

Electricity access

map of “dark” areas crossed with connection registers

Experimental regional GDP

compare γ = 1 and γ = 0.5 with existing regional accounts

Questions to frame your project

- Which economic question?
- Which statistical product, for which user?
- Which official validation data?
- Which confidence level is targeted?
- Which limitations to state?

To get started: open the part 3 notebook in Colab, replace ISO3 with your country, add an event to CUSTOM_EVENTS, and click “Run all”.

Thank you — over to your questions

“A measurement is only useful if you can say how far it can be trusted.”

1. **Lights = a witness of activity, not GDP**
2. **Preprocess before interpreting: clouds, flares, season**
3. **Always an interval and a confidence level**
4. **One ISO3 code → a public, traceable dashboard**

African Development Bank – Data Innovation for Africa (DI4A)

Night-time lights in practice · Analysis notebook “NTL hands-on, part 3”

Glossary of acronyms (1/2)

Acronym	Full form	Meaning here
AfDB / BAD	African Development Bank	institution behind DI4A
DI4A	Data Innovation for Africa	AfDB data-innovation programme
NTL	Night-Time Lights	night-time lights
VIIRS	Visible Infrared Imaging Radiometer Suite	sensor on Suomi-NPP and NOAA-20
DNB	Day/Night Band	VIIRS band sensitive to faint light
DMSP-OLS	Defense Meteorological Satellite Program – Operational Linescan System	older sensor (1992–2013)
VNL V2	VIIRS Nighttime Lights, version 2	EOG annual composites
VNP46A1–A4	NASA Black Marble products	A2 daily, A3 monthly, A4 annual
EOG	Earth Observation Group (Colorado School of Mines)	producer of VIIRS composites
BRDF	Bidirectional Reflectance Distribution Function	viewing-angle correction
GEE	Google Earth Engine	geospatial computing platform
ISO3	ISO 3166-1 alpha-3	three-letter country code
ADM0 / 1 / 2	administrative levels	country / regions / districts
SoL	Sum of Lights	sum of a unit's radiances
cf_cvg	cloud-free coverage	number of cloud-free observations
GHSL	Global Human Settlement Layer (JRC)	population (GHS-POP), urbanisation (GHS-SMOD)

Glossary of acronyms (2/2)

Acronym	Full form	Meaning here
S5P · NO_2	Sentinel-5P · nitrogen dioxide	industrial activity indicator
WDI	World Development Indicators	World Bank database (GDP, access)
PIB / GDP	Gross Domestic Product	constant 2015 USD here
STL	Seasonal-Trend decomposition using Loess	trend + season + remainder
Loess	LOcally Estimated Scatterplot Smoothing	smoothing by local regressions
MCO / OLS	Ordinary Least Squares	classic linear regression
HAC	Heteroskedasticity and Autocorrelation Consistent	Newey–West standard errors
IC / CI	Confidence Interval	95% here
MAD	Median Absolute Deviation	robust dispersion
BIC	Bayesian Information Criterion	penalty against over-segmentation
HHI	Herfindahl–Hirschman Index	concentration index
LOO	Leave-One-Out	validation leaving one year out
ACLED	Armed Conflict Location & Event Data	conflict event database
EM-DAT	Emergency Events Database	disaster database
GAUL	Global Administrative Unit Layers (FAO)	administrative boundaries (fallback)
PAT	Personal Access Token (GitHub)	access token used to publish
LED	Light-Emitting Diode	lighting less visible to VIIRS

References and image sources

- Chen, X., Nordhaus, W. D. (2011). Using luminosity data as a proxy for economic statistics. PNAS 108(21).
- Henderson, J. V., Storeygard, A., Weil, D. N. (2012). Measuring economic growth from outer space. AER 102(2).
- Elvidge, C. D. et al. (2017). VIIRS night-time lights. Int. J. Remote Sensing 38(21).
- Elvidge, C. D. et al. (2021). Annual time series of global VIIRS nighttime lights. Remote Sensing 13(5).
- Román, M. O. et al. (2018). NASA's Black Marble nighttime lights product suite. Remote Sensing of Environment 210.
- Li, X. et al. (2020). A harmonized global nighttime light dataset 1992–2018. Scientific Data 7, 168.
- Gibson, J. et al. (2021). Which night lights data should we use in economics, and where? JDE 149.
- Cleveland, R. B. et al. (1990). STL: a seasonal-trend decomposition procedure based on Loess. JOS 6(1).
- Schiavina, M. et al. (2023). GHS-POP R2023A. European Commission, JRC · Runfola, D. et al. (2020). geoBoundaries. PLoS ONE 15(4).

Image sources

Real images

• NASA Black Marble (Ghana, Switzerland) via blackmarblepy, World Bank — MIT licence

• Mekong, Chicago, platforms: Open Nighttime Lights, World Bank — CC BY 4.0 (after Elvidge et al.; NASA Goddard)

Teaching figures

• generated from simulated data, for illustration only; they describe no real country

Methods and thresholds

• NTL hands-on part 3 notebook and src/ package (config.py)

Cheat sheet: the metrics at a glance

SoL

Σr_i

activity index; also reason per capita

STL

$\log SoL = T + S + R$

read the seasonally adjusted series

Theil–Sen

median of slopes

robust growth; class if ≥ 1%/yr and p < 0.10

β-convergence

$g = a + \beta \log x_0$

β < 0: catching up

HHI

Σs_i^2

concentration; 1 = a single unit

Elasticity

$\log NTL = \alpha + \beta \log GDP$

compare levels and growth; read the CI

Nowcast

$\Delta \log GDP = a + \gamma \Delta \log NTL$

experimental statistic, 80% interval

Anomaly

$|z| \geq 3.5$; ≥ 5 %

robust z (median, MAD) on the STL remainder

Trough

max. gap / 12 months before

+ months to recovery

Gini

A / (A + B)

0 equality · 1 all in one unit

Theil T

between + within

at which level to act?

In the dark

pop. ≥ 5 · r < 0.5 nW

publish a range (0.25–1 nW)